

ASCENSION, Saint Helena

WMO: 619020

Lat: 7.9703S

Lon: 14.4055W

Elev: 79

StdP: 100.38

Time Zone: 0.00 (E00)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	21.1	21.6	14.5	10.4	24.6	15.1	10.8	24.5	11.5	23.9	10.5	24.2	6.5	120	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	4.1	30.2	24.0	30.0	23.9	29.6	23.7	25.1	28.5	24.7	28.2	24.4	28.0	8.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.2	27.0	23.7	18.7	27.0	23.2	18.1	27.0	76.6	28.7	75.4	28.2	74.2	27.9	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	10.1	9.5	DB	20.0	31.2	1.0	0.5	19.3	31.6	18.7	31.9	18.1	32.2	17.4	32.6	(4)
(5)			WB	16.4	26.3	0.7	0.8	15.9	26.8	15.5	27.3	15.1	27.7	14.6	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	26.0	27.0	27.7	27.1	26.1	25.1	24.3	24.0	24.2	24.5	25.2	(6)	
(7)		DBStd	1.47	0.59	0.58	0.63	0.58	0.60	0.62	0.68	0.75	0.66	0.64	0.62	0.57	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5743	497	477	549	531	530	482	469	443	419	439	434	471	(10)
(11)		CDD18.3	2702	239	243	291	281	272	232	211	185	169	181	184	212	(11)
(12)		CDH23.3	18216	1666	2161	2920	2832	2482	1674	1078	627	470	558	666	1083	(12)
(13)	CDH26.7	2702	207	419	698	637	425	160	49	10	4	9	18	66	(13)	
(14)	Wind	WSAvg	7.0	6.6	6.6	6.8	7.2	7.3	7.4	7.2	7.0	6.9	7.1	6.9	(14)	
(15)	Precipitation	PrecAvg	117	6	5	14	24	8	12	9	10	7	6	5	(15)	
(16)		PrecMax	242	18	20	53	97	29	34	34	28	30	33	22	18	(16)
(17)		PrecMin	28	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	60	5	5	14	28	8	9	8	7	6	7	5	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.3	30.2	31.1	30.8	30.1	29.1	28.2	27.3	27.2	27.9	28.2	(19)	
(20)			MCWB	23.1	23.9	24.4	24.3	23.5	22.2	21.2	20.7	20.6	20.7	21.1	21.9	(20)
(21)		2%	DB	28.9	29.9	30.3	30.2	29.6	28.4	27.7	27.0	26.6	26.9	27.1	27.9	(21)
(22)			MCWB	22.8	23.8	24.1	24.0	23.3	22.0	21.1	20.6	20.4	20.6	21.0	21.9	(22)
(23)		5%	DB	28.2	29.2	30.0	29.9	29.1	28.1	27.2	26.5	26.1	26.2	26.8	27.3	(23)
(24)			MCWB	22.6	23.4	24.0	24.0	23.1	22.0	21.0	20.5	20.3	20.5	20.9	21.7	(24)
(25)		10%	DB	27.9	28.9	29.6	29.4	28.8	27.8	26.8	26.0	25.7	25.9	26.1	27.0	(25)
(26)			MCWB	22.4	23.3	23.9	23.8	23.0	21.9	20.9	20.4	20.3	20.4	20.9	21.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.3	25.2	25.6	25.6	24.8	24.0	22.9	22.4	22.2	22.6	23.4	(27)	
(28)			MADB	27.3	28.5	29.2	28.8	28.0	27.3	25.9	25.1	25.2	25.5	25.5	26.5	(28)
(29)		2%	WB	23.7	24.6	25.1	25.1	24.3	23.3	22.3	21.8	21.7	21.7	22.1	22.8	(29)
(30)			MADB	27.1	28.0	28.7	28.4	27.8	26.8	25.7	24.9	24.7	24.9	25.3	26.1	(30)
(31)		5%	WB	23.4	24.2	24.7	24.7	24.0	22.9	22.0	21.4	21.2	21.4	21.8	22.5	(31)
(32)			MADB	26.8	27.7	28.4	28.2	27.6	26.5	25.6	24.6	24.2	24.6	25.0	25.8	(32)
(33)		10%	WB	23.0	23.9	24.4	24.4	23.7	22.5	21.6	21.1	21.0	21.1	21.5	22.1	(33)
(34)			MADB	26.6	27.5	28.1	28.0	27.4	26.3	25.3	24.4	24.0	24.3	24.7	25.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	4.5	4.5	4.1	3.9	3.8	3.9	4.0	4.1	4.2	4.2	4.2	(35)
(36)			MCDBR	4.7	4.8	4.7	4.3	4.1	4.0	4.1	4.2	4.4	4.4	4.6	4.5	(36)
(37)			MCWBR	1.8	1.8	1.8	1.7	1.8	1.9	2.1	2.0	2.1	2.0	2.0	1.9	(37)
(38)		5% WB	MCDBR	4.5	4.7	4.5	4.1	3.9	3.7	3.9	4.0	4.1	4.3	4.2	4.3	(38)
(39)			MCWBR	1.9	1.9	1.9	1.8	1.9	2.1	2.2	2.1	2.2	2.1	2.0	1.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.419	0.423	0.414	0.404	0.392	0.391	0.409	0.460	0.493	0.442	0.396	0.412	(40)	
(41)		taud	2.435	2.435	2.495	2.526	2.521	2.490	2.427	2.249	2.127	2.313	2.495	2.448	(41)	
(42)		Ebn at Noon	922	914	909	891	872	858	848	827	826	886	937	925	(42)	
(43)		Edh at Noon	122	122	113	105	101	101	109	137	161	136	114	120	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.86	7.07	6.91	6.29	5.72	5.34	5.28	5.28	5.50	5.62	5.82	6.32	(44)	
(45)		RadStd	0.17	0.18	0.17	0.20	0.14	0.15	0.12	0.25	0.28	0.21	0.25	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page